Physical Jacobi Metrics and Quantitative Defects in the Metric Blaschke Comparison

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Abstract

The assigned metric Blaschke manuscript compares a full Jacobi determinant with a rank-one symmetric model by a fixed Lagrangian splitting and a weighted logarithmic cancellation. We preserve its physical velocity metric, four endpoint orders, cut-locus poles, and global topological dependencies. The local refinement retains the positive endpoint Schur remainders instead of discarding them. Their relative log-determinants, two families of scalar Jensen gaps, and a density divergence fit simultaneously inside the same geodesic volume defect. This gives a conditional measured matrix-stability statement and a ratio-free density bound $\|f - w\|_1 \leq \sqrt{\delta(4W + \delta)}$. Explicit noncommuting polynomial curves, interior Jacobi residual estimates, and a resonant model explain what finite boundary certificates can and cannot establish. No new classification theorem, curvature stability theorem, or global proof audit is asserted. Contribution assessment: C3/5, a source-specific retained matrix-defect inequality and quantitative density stability, with endpoint geometry and global classification inputs explicitly preserved.

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1 Problem, status, and proof dependencies

The source [1] asserts that every connected closed smooth positive-dimensional Riemannian manifold satisfying

$$\operatorname{inj}(M, g) = \operatorname{diam}(M, g)$$

is a compact rank-one symmetric space up to scale. This report does not adopt that headline as an independently certified conclusion. Its analysis concerns the Jacobi comparison module and quantitative information that this module retains before the source's global upper-volume argument is used.

Write $D = \pi/2$, $T = 2D = \pi$, $d = n - 1$, and let k be the cut-locus nullity. The intermediate case is $0 < k < d$; put $m = d - k$. The source imports classical Blaschke structure: prime simply closed geodesics of length T, great-sphere cut fibers of dimension k, and the associated cohomology model. The density of that model in these units is

$$w(t) = \sin^m t \left|\frac{\sin 2t}{2}\right|^k, \quad 0 \le t \le T, \qquad V_0 = \sigma_d \int_0^D w(t)\,\mathrm{d}t, \tag{1}$$

where σ_d is the area of the unit d-sphere. The parameters for complex, quaternionic, and Cayley type are respectively $(n, k) = (2r, 1), (4r, 3), (16, 7)$. Curvature eigenvalues 1 and 4 describe the model, not a pointwise curvature splitting assumed for M. The sphere and real-projective cases and dimension one are separate source inputs, not cases obtained by dividing our formulas by $k = 0$ or $m = 0$.

Source modules	Mechanism and boundary of this report
Geometry; Lagrangian	Cut-time structure, physical velocity form, pole subtraction, four Schur paths. Selected local identities are rederived below.
Weighted; global comparison	Reflection-weighted Jensen cancellation; Liouville integration; equality gives radial density. The quantitative refinement is located here.
Incidence; signature	Actual cut-normal bundles, integral characteristic classes, quaternionic Pontryagin formulas and signature restrictions. Read, but imported.
Geodesic space; volume algebra	Periodic index domain, rank-three negative bundle, primitive attachment, middle intersection determinant and contact-volume upper bound. Read, but not independently reconstructed or validated.
Classification	Complex homotopy/volume input, projective-plane classification and weak Blaschke volume input, quaternionic upper bound, harmonic rigidity. These remain global obligations.

The primary publisher page identifies [1] as a September 23, 2026 OpenAI Math Release preprint. The October 9 public refresh also inspected McKay's corrected survey v5, dated November 15, 2016 [2]. Its corrected status table and warning about earlier flawed proofs are important reasons not to infer metric classification from an old topological summary. This report does not identify the distinct external results used by the assigned source merely by author name or treat that warning as a disproof of the assigned manuscript. Kreyssig's primary abstract [3] describes the compact simply connected harmonic classification route; that external proof was not reread here. Literature access is scoped, not an exhaustive priority search or a claim of independent acceptance of the new preprint.

2 The physical matrix problem

2.1 Wronskian and normalization

Along a fixed unit geodesic ξ, normal Jacobi fields solve $J'' + R_t J = 0$ with self-adjoint curvature endomorphism R_t. Their $2d$-dimensional solution space has the constant symplectic form

$$\omega(I, J) = \langle I, J' \rangle - \langle I', J \rangle.$$

Let $L_t = \{J : J(t) = 0\}$. Its positive velocity form is $G_t(I, J) = \langle I'(t), J'(t) \rangle$ on L_t. Use one orthonormal normal basis at time zero for both initial values and initial derivatives. On $\mathcal{I} = (0, D) \cup (D, T)$, write L_t as $p = A(t)q$. The source's sign convention gives

$$A = A^{\mathsf{T}}, \qquad A' > 0, \qquad f_\xi(x, y) = \frac{|\det(A(y) - A(x))|}{\sqrt{\det A'(x) \det A'(y)}}. \tag{2}$$

Indeed, the graph frame $E_t q = (q, A(t)q)$ has Gram matrix $A'(t)$ for G_t. Multiplication by $A'(t)^{-1/2}$ makes that frame physically orthonormal, and the cross-Wronskian is the normalized difference $A(y) - A(x)$. This is a congruence computation, not a replacement of a physical norm by a convenient coefficient norm.

The continuous extension of f_ξ is symmetric and separately T-periodic; it is the absolute polar Jacobi determinant. In polar integration there is no additional factor t^d multiplying this f. The Cartesian density would instead be $f_\xi(0, t)/t^d$. The periodicity of Jacobi fields in the tangent bundle follows from the return map of geodesic flow. It does *not* say that a parallel normal frame has trivial holonomy.

2.2 Fixed blocks and their poles

Choose the first k initial derivative coordinates to span $L_0 \cap L_D$, and retain this fixed splitting throughout:

$$A = \begin{pmatrix} V & C \\ C^{\mathsf{T}} & H \end{pmatrix}, \qquad S = V' - C' H'^{-1} C'^{\mathsf{T}} > 0, \qquad H' > 0. \tag{3}$$

Here V is k by k, H is m by m, and C is k by m. In particular $\det A' = \det H' \det S$; simultaneous diagonalization of these matrices is neither available nor needed.

The actual source frame analysis gives

$$\begin{aligned} A(t) &= -\frac{I_d}{t - t_0} + \text{smooth remainder}, && t_0 = 0, T, \\ A(t) &= -\frac{\operatorname{diag}(B, 0)}{t - D} + \text{smooth remainder}, && B > 0. \end{aligned} \tag{4}$$

The middle pole is not obtained by projecting a scalar Riccati equation. In a smooth Jacobi frame (X, P), divide the k vanishing value columns of X by $t - D$; the limiting matrix is invertible by the physical velocity form. The Wronskian symmetry identifies the residue as $-\operatorname{diag}(B, 0)$ with $B > 0$. Thus C, H extend smoothly through D and $H'(D) > 0$.

Set

$$h = (\det H')^{1/m}, \quad s = (\det S)^{1/k}, \quad s_i(t) = s(t + iD), \quad i = 0, 1. \tag{5}$$

On a fixed geodesic, $h(t) \sin^2 t$ and $s_i(t) \sin^2(2t)/4$ are bounded above and below by positive constants on the closed appropriate intervals. At the endpoints these follow from the pole residues and $H', S \sim (t - t_0)^{-2}$ times positive matrices. The constants are not supplied uniformly over arbitrary metrics. No integral of H' or S across a pole is used below.

3 Retaining the matrix coupling defects

For $x \neq y$ in $\mathcal{I}$, define

$$\Delta H = H(y) - H(x), \quad \Delta C = C(y) - C(x), \quad Q(x,y) = V(y) - V(x) - \Delta C\, \Delta H^{-1} \Delta C^{\mathsf{T}}.$$

H is strictly increasing across $(0,T)$, so ΔH is invertible also when $x > y$. Let $R(x,y) = \Delta C\, \Delta H^{-1}$ and $Z(t) = C'(t)H'(t)^{-1}$. Differentiating the inverse in the displayed formula, and completing a matrix square, gives the source identities

$$\begin{aligned} \partial_y Q(x,y) &= S(y) + (R(x,y) - Z(y))H'(y)(R(x,y) - Z(y))^{\mathsf{T}}, \\ -\partial_x Q(x,y) &= S(x) + (R(x,y) - Z(x))H'(x)(R(x,y) - Z(x))^{\mathsf{T}}. \end{aligned} \tag{6}$$

The order of all factors matters. These formulas do not require $\Delta H > 0$, only its invertibility.

For $0 < u < v < D$ and $i, j \in \{0,1\}$ write $Q_{ij} = Q(u + iD, v + jD)$, and set

$$B_a(u,v) = \int_u^v S(t + aD)\, \mathrm{d}t, \qquad I_a(u,v) = \int_u^v s_a(t)\, \mathrm{d}t, \quad a = 0, 1.$$

Both are finite on each such interior interval.

Proposition 3.1 (Two positive defect representations)**.** *Under the source's diagonal and cut-intersection identities, $Q_{ij} = B_j + \Omega_{ij}^{[y]} = B_i + \Omega_{ij}^{[x]}$, where*

$$\begin{aligned} \Omega_{ij}^{[y]} &= \int_u^v [R(u + iD, t + jD) - Z(t + jD)]H'(t + jD)[R(u + iD, t + jD) - Z(t + jD)]^{\mathsf{T}}\, \mathrm{d}t, \\ \Omega_{ij}^{[x]} &= \int_u^v [R(t + iD, v + jD) - Z(t + iD)]H'(t + iD)[R(t + iD, v + jD) - Z(t + iD)]^{\mathsf{T}}\, \mathrm{d}t. \end{aligned} \tag{7}$$

In particular each Ω is positive semidefinite and $Q_{ij} > 0$, including $(i,j) = (1,0)$.

Proof. At equal times, Taylor expansion gives $Q(x,y) = (y-x)S(x) + O((y-x)^2)$, hence the diagonal extension is zero. At separation D, the full difference $A(y) - A(x)$ has kernel dimension k, whereas ΔH is invertible. Its k by k Schur complement therefore has full kernel and is zero. These are the starting values for the two paths: integrate $\partial_y Q$ from $u + jD$ to $v + jD$, or integrate $-\partial_x Q$ backwards from $v + iD$ to $u + iD$. Each path stays in one half-period; in the reversed endpoint case the starting value is the reverse cut pair, still zero. Equation (6) proves the formulas. At a diagonal endpoint R has its continuous limiting value Z, so no artificial singularity is inserted. □

Define the dimensionless relative defects and their logarithms by

$$\begin{aligned} E_{ij}^{[y]} &= B_j^{-1/2} \Omega_{ij}^{[y]} B_j^{-1/2}, & \rho_{ij}^{[y]} &= \log\det(I_k + E_{ij}^{[y]}), \\ E_{ij}^{[x]} &= B_i^{-1/2} \Omega_{ij}^{[x]} B_i^{-1/2}, & \rho_{ij}^{[x]} &= \log\det(I_k + E_{ij}^{[x]}). \end{aligned} \tag{8}$$

They are nonnegative. Their eigenvalues and determinant ratios are invariant under a consistent invertible change of block coordinates by congruence; their entrywise sizes in an arbitrary chart are not physical invariants. For either representation,

$$\det Q_{ij} = \det B_a\, e^{\rho}, \qquad (\det B_a)^{1/k} \geq I_a.$$

The last inequality is the classical Minkowski determinant inequality for an integral of positive matrices. It follows first for positive Riemann sums from the concavity and homogeneity of $\det^{1/k}$, and then by continuity. We credit this classical tool, not a new determinant inequality.

The determinant in (2) factors as $f = f_H f_K$, with

$$f_H(x,y) = \frac{|\det \Delta H|}{\sqrt{\det H'(x) \det H'(y)}}, \qquad f_K(x,y) = \frac{|\det Q(x,y)|}{\sqrt{\det S(x) \det S(y)}}.$$

Put $b_0(z) = \sin(2z)/2$ for $0 < z < D$, and

$$g_H(t) = \log(h(t)\sin^2 t), \qquad g_a(t) = \log(s_a(t) b_0(t)^2).$$

These functions are bounded, by (4). Writing $I_*(u,v) = \int_u^v 4\csc^2(2t)\,\mathrm{d}t$ and $A_a(u,v) = \log(I_a/I_*)$, the averaged two path estimates give

$$\log \frac{f_K(u+iD, v+jD)}{b_0(v-u)^k} \geq \frac{k}{2}\{A_i + A_j - g_i(u) - g_j(v)\} + \frac{\rho_{ij}^{[x]} + \rho_{ij}^{[y]}}{2}. \tag{9}$$

This is the specific extra term retained here. The source already proves the matrix squares and the inequality obtained by deleting the last term. No source error or new Schur principle is claimed.

4 A quantitative weighted comparison

4.1 The cancellation with its exact measure

Let $b > 0$, $\lambda = \pi/b$, $p_0(t) = \lambda^2 \csc^2(\lambda t)$, and $\Delta_b = \{0 < x < y < b\}$. For each interior interval define

$$d\nu_{xy}(t) = \frac{p_0(t)\,dt}{\int_x^y p_0}, \qquad \mathcal{J}_{xy}(g) = \log \int e^g \, d\nu_{xy} - \int g \, d\nu_{xy} \geq 0.$$

If $p = p_0 e^g$ with bounded g, and $W_b(z) = W_b(b-z) \geq 0$ with $W_b(z)/\sin(\lambda z)$ bounded, then

$$\int_{\Delta_b} W_b(y-x) \left[\log \frac{\int_x^y p}{\int_x^y p_0} - \frac{g(x)+g(y)}{2} \right] \mathrm{d}x\,\mathrm{d}y = \int_{\Delta_b} W_b(y-x) \mathcal{J}_{xy}(g)\,\mathrm{d}x\,\mathrm{d}y. \tag{10}$$

Here and below no unweighted integral of p_0 on the whole interval is asserted finite.

For completeness, $\int_x^y p_0 = \lambda \sin(\lambda(y-x))/(\sin\lambda x \sin\lambda y)$. Fubini for bounded g reduces (10) to the coefficient of $g(t)$. With $K(z) = W_b(z)/\sin(\lambda z)$, that coefficient in the interval average is

$$\frac{p_0(t)}{\lambda} \int_{x<t<y} K(y-x)\sin(\lambda x)\sin(\lambda y)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{2}\int_0^b W_b(z)\,\mathrm{d}z.$$

The inner integral is $\sin^2(\lambda t)(\int_0^b W_b)/(2\lambda)$: split the $z = y - x$ integral at $t, b-t$, integrate the two sine factors, and pair the terms z and $b-z$. The same constant is the coefficient of $(g(x)+g(y))/2$, since reflection makes the two endpoint integrals sum to $\int_0^b W_b$. This proves the identity and justifies all interchanges. Nonnegative measurable weights follow by bounded approximation. This is the source cancellation, restated with its probability measure explicit.

Let $\mathcal{R} = (0,T)^2$, write $w_{xy} = w(|y-x|)$, and set

$$W_* = \int_{\mathcal{R}} w_{xy}\,\mathrm{d}x\,\mathrm{d}y, \qquad \delta_\xi = \int_{\mathcal{R}} (f_\xi - w_{xy})\,\mathrm{d}x\,\mathrm{d}y.$$

For folded pairs, $z = v - u$,

$$w_{ij}(z) = b_0(z)^k \begin{cases} \sin^m z & i = j, \\ \cos^m z & i \neq j, \end{cases} \qquad W_D(z) = (\sin^m z + \cos^m z) b_0(z)^k. \tag{11}$$

Every row and every column of (w_{ij}) sums to W_D. The weight w satisfies the hypotheses of (10) on $(0,T)$; W_D does so on $(0,D)$, including $k = 1$.

Theorem 4.1 (Joint density and matrix defect budget)**.** *Assume the actual physical Jacobi setup, poles, and cut identities above. With $\Phi(r) = r - 1 - \log r$ for $r > 0$, one has*

$$\begin{aligned}\delta_\xi \;&\geq \int_{\mathcal{R}} w_{xy}\Phi(f_\xi/w_{xy})\,\mathrm{d}x\,\mathrm{d}y + 2m\int_{\Delta_T} w(y-x)\mathcal{J}_{xy}(g_H)\,\mathrm{d}x\,\mathrm{d}y \\ &\quad + 2k\sum_{a=0}^{1}\int_{\Delta_D} W_D(v-u)\mathcal{J}_{uv}(g_a)\,\mathrm{d}u\,\mathrm{d}v \\ &\quad + \sum_{i,j=0}^{1}\int_{\Delta_D} w_{ij}(v-u)(\rho_{ij}^{[x]} + \rho_{ij}^{[y]})\,\mathrm{d}u\,\mathrm{d}v.\end{aligned} \tag{12}$$

The interval probability in the second term uses $p_0 = \csc^2 t$; in the third it uses $p_0 = 4\csc^2(2t)$. Every term on the right is nonnegative and finite.

Proof. Minkowski applied to H' bounds $f_H^{1/m}$ below by $|\int_x^y h|/\sqrt{h(x)h(y)}$. Taking logarithms and applying (10), then using symmetry, bounds the full-square H logarithm below by the second term in (12).

For K, split the square, outside null lines, into the four folded regions $(u + iD, v + jD)$ and their four transposes. On each first region use (9); symmetry gives the factor two for its transpose. The row and column sums in (11) leave precisely $2k\sum_a \int W_D[A_a - (g_a(u) + g_a(v))/2]$. Use (10) separately for $a = 0, 1$. The remaining matrix terms have coefficient one, exactly as displayed.

Thus $\int_{\mathcal{R}} w\log(f/w)$ dominates the last three terms. The identity $f - w = w\log(f/w) + w\Phi(f/w)$ completes the proof. There is no hidden endpoint integrability assumption: the scalar logarithmic lower bounds are bounded below because the g's are bounded. Also $w(\log(f/w))_+ \leq f$, and f is continuous on the compact square. Consequently the weighted logarithms are integrable. The nonnegative retained terms are then integrable by the very inequality just proved, using exhaustion if necessary. □

Remark 4.2. Equation (12) is a conditional quantitative strengthening of the source comparison, not a hypothesis-free estimate for arbitrary positive matrix curves. In particular it uses the two cut anchors $Q(u, u + D) = 0$ and the precise reflection weights. Bounded interior matrix residuals cannot manufacture either premise.

5 Density stability without a ratio bound

The scalar consequences need only finite nonnegative densities f, w on a measure space, $w > 0$ almost everywhere, $W = \int w > 0$, $F = \int f = W + \delta$, and $\int w\log(f/w) \geq 0$. These conditions follow on the geodesic square from the preceding theorem. The zero sets of the model weight are null lines and cause no difficulty.

Theorem 5.1 (Two complementary density bounds)**.** *Under these conditions $\delta \geq 0$ and*

$$H^2 := \int(\sqrt{f} - \sqrt{w})^2 \leq \delta, \tag{13}$$

$$\|f - w\|_1 \leq \min\left\{\sqrt{\delta(4W + \delta)},\ \delta + W\sqrt{2\log(1 + \delta/W)}\right\}. \tag{14}$$

More generally, if $0 \leq L \leq \int w\log(f/w)$, then

$$H^2 \leq \delta - L, \qquad \|f - w\|_1 \leq \sqrt{(\delta - L)(4W + \delta + L)}. \tag{15}$$

Thus a certified positive matrix/Jensen reserve in (12) *can be allocated before bounding density error.*

Proof. For $r > 0$,

$$\Phi(r) - (\sqrt{r} - 1)^2 = 2(\sqrt{r} - 1 - \log\sqrt{r}) \geq 0.$$

The identity for δ used above proves $H^2 \leq \delta - L$. Cauchy–Schwarz, with the exact sum of squares rather than its upper bound, gives

$$\|f - w\|_1^2 \leq H^2 \int (\sqrt{f} + \sqrt{w})^2 = H^2(4W + 2\delta - H^2).$$

The quadratic on the right increases for $0 \leq H^2 \leq \delta$. This proves (15) and the first bound in (14). Equivalently Jensen gives the affinity $\int \sqrt{fw} \geq W$; this explains the coefficient $4W + \delta$ instead of the looser $4W + 2\delta$.

For the second bound normalize $p = w/W$ and $q = f/F$. Their relative entropy, in this order, satisfies

$$\int p \log(p/q) = \log(F/W) - \frac{1}{W} \int w \log(f/w) \leq \log(1 + \delta/W).$$

For clarity, the classical Pinsker bound used here follows by splitting at $\{p > q\}$, applying Jensen on each part (the log-sum inequality), and reducing to Bernoulli relative entropy. For fixed Bernoulli q, the second derivative in its first parameter x is $1/[x(1-x)] \geq 4$. The value and first derivative vanish at $x = q$, so the entropy is at least twice the squared difference of the two event probabilities, namely $\|p - q\|_1^2/2$. The triangle inequality $\|f - w\|_1 \leq W\|q - p\|_1 + \delta$ proves the result. Approximation handles null sets and endpoint Bernoulli parameters. □

These are classical convexity and information inequalities deployed at the source's actual weighted comparison interface. Their small-gap constants differ: the Pinsker expression is $\sqrt{2W\delta} + O(\delta)$, whereas the affinity expression is $2\sqrt{W\delta} + O(\delta^{3/2})$. Neither is asserted to be the optimal density modulus under every additional Jacobi constraint. When $\delta = 0$, $f = w$ almost everywhere and hence everywhere if both are continuous; that equality conclusion is already in the source.

Corollary 5.2 (A measured, not uniform, geometric consequence)**.** *Assume in addition the source's polar-volume and Liouville identities on the actual Blaschke manifold. Let* $V = \mathrm{Vol}(M, g)$ *and* $\zeta = (V - V_0)/V_0$. *For the probability measure*

$$d\mathbb{P}(\xi, x, y) = \frac{w(|y - x|)\, d\mu(\xi)\, dx\, dy}{2TVV_0}, \qquad d\mu = d\,\mathrm{Vol}_g\, d\sigma, \tag{16}$$

put $\Psi = (\rho_{ij}^{[x]} + \rho_{ij}^{[y]})/2$ *on a folded region and give its transpose the same value. Then*

$$\mathbb{E}_{\mathbb{P}}[\Phi(f_\xi/w) + \Psi] \leq \zeta, \tag{17}$$

$$\mathbb{P}\Big(\max\{\|E_{ij}^{[x]}\|_{\mathrm{op}}, \|E_{ij}^{[y]}\|_{\mathrm{op}}\} \geq \eta\Big) \leq \min\left\{1, \frac{2\zeta}{\log(1+\eta)}\right\}, \qquad \eta > 0. \tag{18}$$

Moreover, with the ratio interpreted off the null weight lines,

$$\mathbb{E}_{\mathbb{P}}|f_\xi/w - 1| \leq \min\{\sqrt{\zeta(4+\zeta)},\ \zeta + \sqrt{2\log(1+\zeta)}\}.$$

Proof. The source normalizations give

$$\int_{SM} \int_{\mathcal{R}} f_\xi = 2TV^2, \qquad \int_{SM} \int_{\mathcal{R}} w = 2TVV_0.$$

One way to see the first identity is to write $f_\xi(x, y) = J_{\phi_x \xi}((y - x) \bmod T)$, use invariance of Liouville measure, and use the two half-period polar integrals. The model identity is identical with J replaced by w. Integrate (12) and divide by $2TVV_0$; the folded matrix sum is exactly $\int w\Psi$, not twice that integral. For a positive semidefinite matrix E, $\log\det(I + E) \geq \log(1 + \|E\|_{\mathrm{op}})$. Markov's inequality proves (18). Apply Theorem 5.1 to densities f_ξ/w and 1 on this probability space for the last assertion. □

This statement uses neither the source's volume upper bound nor its final classification. If that entire classification were assumed, then $\zeta = 0$ and these estimates would simply recover zero defects. Their role is quantitative information in the comparison module before that final implication. No uniform bound on every geodesic, curvature tensor, injectivity radius of a perturbation, or Gromov–Hausdorff distance is proved.

6 Explicit noncommuting curves and sharp limitations

Example 6.1 (A positive curve with a genuine discarded term)**.** Let S_0, H_0 be positive definite matrices, B a rectangular matrix, and on an ordinary interior interval set

$$H(t) = tH_0, \quad C(t) = \tfrac{1}{2}t^2 BH_0, \quad V(t) = tS_0 + \tfrac{1}{3}t^3 BH_0B^{\mathsf{T}}.$$

Its block derivative has Schur complement S_0, hence $A'(t) > 0$. For $a < b$, $\ell = b - a$, direct ordered multiplication gives

$$Q(a,b) = \ell S_0 + \frac{\ell^3}{12} BH_0B^{\mathsf{T}}. \tag{19}$$

Indeed $\Delta C\, \Delta H^{-1} = (a+b)B/2$, and the two square integrals in (6) both equal $\int_a^b (t-(a+b)/2)^2 BH_0B^{\mathsf{T}}dt$. For

$$H_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad S_0 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the coupling matrix is $\left(\begin{smallmatrix} 6 & 3 \\ 3 & 2 \end{smallmatrix}\right)$, which does not commute with S_0. Nevertheless

$$\frac{\det Q(a,b)}{\det(\ell S_0)} = 1 + \frac{7\ell^2}{12} + \frac{\ell^4}{96}; \qquad \ell = 1: \quad \rho = \log(51/32) > 0. \tag{20}$$

Deleting the remainder loses precisely this positive log-determinant in the determinant step. This is an exact positive Lagrangian graph curve, not a constructed closed Blaschke metric or a fixture for its cut anchors.

Example 6.2 (A global matrix tail is not a trace bound)**.** On an abstract probability space let $E_N = N$ on a set of probability $(\log(1+N))^{-2}$ and zero elsewhere, for N large. Then $\mathbb{E}\log(1+E_N) = 1/\log(1+N) \to 0$, but $\mathbb{E}E_N = N/(\log(1+N))^2 \to \infty$. Thus (18) cannot be upgraded to mean trace control without additional uniform integrability or a spectral ceiling. Under the extra assumption $0 \le E \le LI$,

$$\operatorname{tr} E \le \frac{L}{\log(1+L)} \log\det(I+E),$$

with the limiting convention at $L = 0$. This follows because $x/\log(1+x)$ is increasing. The example does not assert such a concentration pattern is realizable by Blaschke metrics.

Example 6.3 (The model already has a resonant boundary problem)**.** In a model curvature-4 direction, $J(t) = \sin(2t)$ vanishes at $0, D = \pi/2$ and solves $J'' + 4J = 0$. Thus a zero differential and Dirichlet residual does not bound the solution error: arbitrary multiples of J have zero residual. On $H^2(0,D) \cap H_0^1(0,D)$ the operator $-\partial_t^2 - 4$ has this kernel. Solvability requires orthogonality of the forcing to J, and uniqueness requires a gauge. On $(0,b)$, $b < D$, its first eigenvalue is $(\pi/b)^2 - 4$, tending to zero as $b \uparrow D$. This is the exact cut-endpoint obstruction, not a numerical conditioning accident.

7 Bounded interior certificates and endpoint costs

7.1 A Volterra certificate in the actual normal metric

The original boundary-value Ram report [4], especially lines 444–474 and 759–807, distinguishes a local jet pivot from an inverse on a boundary domain. The physical-metric chapter [5] requires the reconstruction Gram form and omitted residual, not just a finite compressed matrix. A useful compatible certificate here is an initial-value estimate on a pole-free interval, not a global Riccati inverse.

Proposition 7.1 (Interior Jacobi error)**.** *Work in a parallel orthonormal frame on* $[0, L]$ *and suppose* $\|R(t)\|_{\mathrm{op}} \le K$ *with* $K \ge 0$*. Let* Y *solve* $Y'' + RY = 0$ *and let a twice weakly differentiable approximate matrix* $\widetilde{Y}$ *have residual* $r = \widetilde{Y}'' + R\widetilde{Y} \in L^1$*. If the initial value and derivative errors have norms at most* a, b*, then, for* $0 \le t \le L$*,*

$$\|Y(t) - \widetilde{Y}(t)\|_{\mathrm{op}} \le a\cosh(\sqrt{K}t) + b\, s_K(t) + \int_0^t s_K(t-u)\|r(u)\|_{\mathrm{op}}\,\mathrm{d}u, \qquad s_K(t) = \frac{\sinh(\sqrt{K}t)}{\sqrt{K}}. \tag{21}$$

At $K = 0$*,* $s_0(t) = t$*. The derivative error is bounded by the derivative in* t *of the right side.*

Proof. Twice integrate the error equation to obtain the scalar inequality $e(t) \le a + bt + \int_0^t (t-u)(Ke(u) + \|r(u)\|)du$. Iterate the positive Volterra kernel. Its jth power has factorial denominators, so its series converges uniformly on $[0, L]$. Summing gives the hyperbolic majorant (21); the remainder vanishes by the same factorial bound. Integrating the first derivative equation and inserting this majorant proves the derivative assertion. No commutation between $R(t)$ and $R(u)$ has been used. □

If instead an approximate curvature $\widetilde{R}$ is used, the residual budget must include $(R - \widetilde{R})\widetilde{Y}$ in this same physical norm. A finite list of samples does not provide the required L^1 or uniform bounds. At a change of parallel frame, the error matrices transform orthogonally. Identifying the two ends of a closed geodesic requires the holonomy map; periodic scalar coefficient rows cannot replace that map.

7.2 Relative determinant enclosures

For a positive definite matrix B and an approximation $\widehat{B}$, assume a certified *relative form* estimate

$$\|B^{-1/2}(\widehat{B} - B)B^{-1/2}\|_{\mathrm{op}} \le \varepsilon_B < 1.$$

Then $(1 - \varepsilon_B)B \preceq \widehat{B} \preceq (1 + \varepsilon_B)B$, so

$$|\log\det\widehat{B} - \log\det B| \le k[-\log(1 - \varepsilon_B)]. \tag{22}$$

Apply the same statement to Q with relative error ε_Q. The computed difference $\log\det\widehat{Q} - \log\det\widehat{B}$ then encloses $\rho = \log\det Q - \log\det B$ with radius $k[-\log(1 - \varepsilon_Q) - \log(1 - \varepsilon_B)]$. The proof is the product of the eigenvalue inequalities for the congruence-normalized matrix; arbitrary small entry errors alone do not imply the relative premise. A negative computed lower bound is replaced by zero only when the exact path positivity has already been established, not to conceal a failed cut constraint.

The endpoint analysis must treat the explicit principal parts (4) separately. After pole subtraction, bounded smooth remainders and their derivatives can be handled on compact panels. Certifying an approximate residue is not the same as proving its exact rank, its positivity, or the geometrically imposed cut intersection. These remain source premises in the comparison theorem.

7.3 Density quadrature without logarithmic endpoint substitution

Suppose an integrable computed density $\widehat{f}$ satisfies $\|f - \widehat{f}\|_1 \le \varepsilon$ on the *whole* square, and a certified enclosure gives $\int(\widehat{f} - w) \le d_+$. Then

$$0 \le \delta \le U := d_+ + \varepsilon, \qquad \|f - w\|_1 \le \sqrt{U(4W + U)}, \qquad \|\widehat{f} - w\|_1 \le \varepsilon + \sqrt{U(4W + U)}.$$

This follows by monotonicity in δ. If $U < 0$, the claimed enclosure and source premises are inconsistent. An interior panel calculation must include a certified omitted-strip L^1 budget in ε. It cannot simply insert approximate values into $\log(f/w)$ near the vanishing weight. To allocate a matrix reserve in (15), one additionally needs lower enclosures of its weighted integral, including interval and integration errors.

8 Source comparison, limitations, and research directions

The earned increment is the *joint* retained-defect estimate, not the Wronskian, Schur identity, Minkowski inequality, Jensen inequality, or Pinsker inequality separately. Its advantage is that physical cross-block coupling, scalar profile variation, and density error cannot all independently spend the same volume defect. It requires no global ratio ceiling on f/w and keeps the cut degeneracies at their correct weights. Its cost is that relative matrix information and whole-domain integration are more demanding than checking a few finite Gram matrices.

The original physical-metric chapter provides the relevant discipline: if J reconstructs a finite coefficient vector into a physical Hilbert space, then $G = J^*J$ is its norm. An orthogonal compressed residual misses the perpendicular component unless that component is explicitly bounded. In this report the exact analogue is the velocity metric A' and the complete path integral (7). Neither a well-conditioned coefficient matrix nor an interior approximate Wronskian proves a true cut anchor.

One substantive sign correction was found in the freshly read private original, not in the assigned public comparison. TR212 lines 960–969 write $-(pu')' = f$, $-p(0)u'(0) = g_0$, $p(1)u'(1) = g_1$, but state $\int f = g_0 + g_1$. For these displayed signs the correct identity is $\int f = -(g_0 + g_1)$, by direct integration. This appendix identity is not used in the present Jacobi proofs; the original file is preserved unchanged. Its resonance and conditional inverse statements are retained with their stated domains. No correction to the assigned source's reversed Schur path or physical determinant normalization is proposed.

The global source modules remain substantial: smooth incidence bundles and their integral classes; signature restrictions; the actual periodic index domain and primitive attachment in free loop space; the contact volume number and intersection-form normalization; the upper-volume algebra; and the external harmonic classification. Reading their proofs does not mean this report independently establishes them. In particular, the half-Jacobi fields used in the source's index construction live in the relevant H^1 form domain; derivative jumps do not make them H^2 operator-domain eigenfunctions. A finite BVP cannot bypass this distinction.

Useful next steps would be a geometric realization theorem for small relative path defects, or additional uniform regularity turning the measured estimate (18) into uniform control. Both need new hypotheses or new proofs. Verified pole-subtracted integration for an actual metric could make the present local bounds computationally effective; no running time, speedup, numerical metric recovery, or optimal geometric stability constant is claimed here. The polynomial example is a test of the noncommutative algebra, not a realization theorem for a compact manifold.

9 Evidence and contribution boundaries

All 15 assigned public text files were displayed literally in bounded untruncated chunks: 3702 lines, including the ten scientific sections, main file, bibliography, two figure sources, and README. The two

private originals were freshly read in full: TR212, 1053 lines, and the physical-metric chapter, 210 lines. Their total is 1263 lines. Pre-read hashes were persisted before these displays; post-read hashes match. Exact intervals, including the confirmation after the connection interruption, are retained in the owned read ledger. The figures' source text was read, not their original rendered PDF. Other cited private reports, the chapter's included figure, the entire private corpus, and full external reference proofs were not read.

The owned test program uses exact rational matrix curves, ordered Schur derivatives, determinant ratios, moment integrals, and negative controls. It passes 47 exact parameter cases with 217 assertions, plus eight 60-digit scalar diagnostic cases with 24 assertions, counted separately. No parent tests or foreign checker execution are counted. These finite checks supplement the written proofs and do not verify cut-locus geometry or the global metric theorem.

Applicability remains L3. The C3 assessment concerns the retained matrix/density quantitative comparison at the actual source interface, with the conditional measured extension and explicit certification costs. It does not credit textbook identities as historically new, nor claim a new solution of the Blaschke conjecture. Full source architecture was read; only the stated local deductions were independently proved here.

References

[1] OpenAI, *The metric Blaschke theorem*, September 23, 2026, OpenAI Math Release preprint. Captured source used in this report: `https://github.com/openai/math/tree/main/preprints/The-metric-Blaschke-theorem-September-23-2026`.

[2] B. McKay, *Summary of progress on the Blaschke conjecture*, arXiv:1309.1326v5, November 15, 2016; corrected version of ICCM Notices 3 (2015), no. 2, 33–45. `https://arxiv.org/abs/1309.1326`. Fresh access: abstract, version history, and HTML Sections 1–4; not a full external proof audit.

[3] P. Kreyssig, *An Introduction to Harmonic Manifolds and the Lichnerowicz Conjecture*, arXiv:1007.0477v1, July 3, 2010. `https://arxiv.org/abs/1007.0477`. Fresh abstract and metadata access only.

[4] M. Subbarao, original private report, *Boundary-value Ram operators*, TR212, local manuscript. Fresh literal read: lines 1–1053; especially resonance 444–474, trace assembly 476–525, and conditional residual bounds 759–807. Local file: `TR212-boundary-value-ram-operators/TR212-boundary-value-ram-operators.tex`.

[5] M. Subbarao, original physical-metric module, beginning *A Nonorthogonal Representation Carries a Metric*, RamCalBooks, volume 21, `chapters/ch01d-physical-metrics.tex`, lines 1–210 freshly read. Physical reconstruction, Gram forms, omitted residuals, and finite-versus-continuum restrictions.